Continuum mechanics – stress, strain and elastic waves

A steel beam, a rubber ball, the Earth's crust and a jar of honey are made of atoms, but on the scales engineers and geophysicists work with, matter can be treated as a continuum: a field of displacements, stresses and velocities. Continuum mechanics is the common language of elasticity theory and fluid mechanics. It builds bridges and aircraft, explains earthquake waves and buckled columns – and is a beautiful example of how tensors arise quite naturally in physics.

The continuum hypothesis

A continuum is an idealisation: one assumes that any small volume dV contains so many atoms that density, velocity and stress are smooth fields – while dV is still small compared with the scales of interest. For a metal beam this works down to micrometres. For a rarefied gas in the upper atmosphere it breaks down when the mean free path becomes comparable with the body (the Knudsen number).

The story begins with Robert Hooke, who in 1676 published the anagram "ceiiinosssttuv" – *ut tensio, sic vis*: as the extension, so the force. Jakob Bernoulli and Leonhard Euler developed the theory of beams in the 18th century. The general theory was created in Paris in the 1820s: Claude-Louis Navier (1821), Augustin-Louis Cauchy (1822–23) and Siméon Poisson. Cauchy's concept of stress is the foundation of everything that follows.

Cauchy's stress tensor

Imagine cutting a body in two. Across the cut acts a force per area – a *traction* $\vec{t}$. Cauchy showed that the traction on a surface with normal $\vec{n}$ depends linearly on $\vec{n}$:

$$t_i = \sigma_{ij}\, n_j, \qquad \sigma = \begin{pmatrix} \sigma_{xx} & \tau_{xy} & \tau_{xz} \\ \tau_{xy} & \sigma_{yy} & \tau_{yz} \\ \tau_{xz} & \tau_{yz} & \sigma_{zz} \end{pmatrix}$$

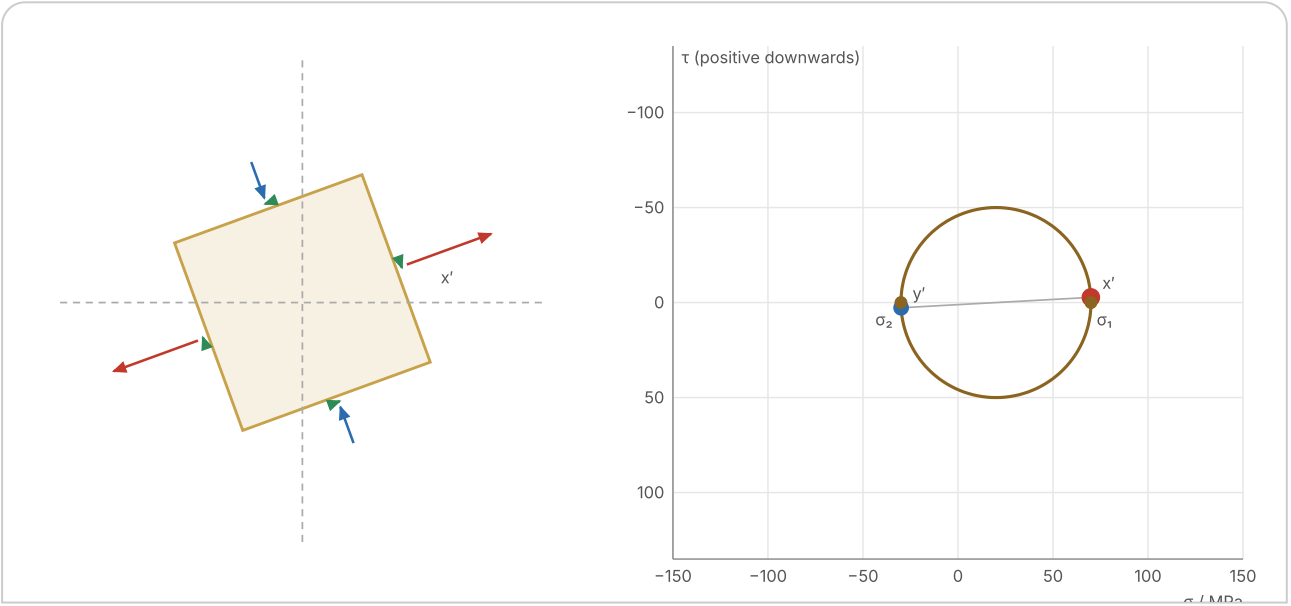
The diagonal elements are normal stresses (tension or compression), the rest shear stresses. The tensor is symmetric – otherwise a small element would start rotating infinitely fast. A pressure p in a fluid at rest corresponds to $\sigma_{ij} = -p\,\delta_{ij}$. The very word "tensor" comes from here: Latin *tendere*, to stretch.

Principal stresses and Mohr’s circle

Because $\boldsymbol{\sigma}$ is symmetric, there are always three orthogonal directions with no shear stress – the principal axes. The eigenvalues are the principal stresses. In two dimensions Christian Otto Mohr showed in 1882 that the stresses on all cut planes lie on a circle in the (σ, τ) plane with centre $\tfrac{1}{2}(\sigma_x + \sigma_y)$ and radius

$$R = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}, \qquad \sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm R, \qquad \tan 2\theta_p = \frac{2\tau_{xy}}{\sigma_x - \sigma_y}$$

The maximum shear stress is R and acts on planes turned 45° from the principal axes. This explains why a stick of chalk breaks in a 45° helix when twisted, while a ductile metal slips along the planes of greatest shear.



Strain

When a body is deformed, the point $\vec{x}$ moves to $\vec{x} + \vec{u}(\vec{x})$. For small deformations the local stretching and change of angle are described by the symmetric strain tensor (the antisymmetric part is merely a rotation):

$$\varepsilon_{ij} = \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right), \qquad \frac{\Delta V}{V} = \varepsilon_{kk} = \nabla \cdot \vec{u}$$

The diagonal elements are relative extensions, the others half the changes of angle. For large deformations – rubber, biological tissue, the Earth’s crust over geological time – non-linear measures of strain (Green–Lagrange) must be used, and then the theory becomes much harder.

Hooke’s law and the elastic moduli

For an isotropic, linearly elastic material there are only two independent constants, the Lamé parameters λ and μ (Gabriel Lamé, 1852):

$$\sigma_{ij} = \lambda \varepsilon_{kk} \delta_{ij} + 2\mu \varepsilon_{ij}$$

In practice the engineering constants are used: Young’s modulus E (Thomas Young, 1807), Poisson’s ratio ν (lateral contraction under tension), the shear modulus $G = \mu$ and the bulk modulus K:

$$G = \frac{E}{2(1+\nu)}, \qquad K = \frac{E}{3(1-2\nu)}, \qquad -1 < \nu < \tfrac{1}{2}$$

Material	E (GPa)	ν	ρ (kg/m³)
Steel	200	0.29	7850
Aluminium	69	0.33	2700
Glass	70	0.22	2500
Concrete	30	0.20	2400
Wood (along the grain)	11	–	500
Bone	18	0.3	1900
Rubber	0.01–0.1	0.4999	1100
Graphene	1000	0.19	–

Rubber is almost incompressible ($\nu \to \tfrac{1}{2}$): it changes shape but not volume. Cork has $\nu \approx 0$, which is why it is good for wine bottles. Auxetic materials with negative ν get thicker when stretched. Crystals are anisotropic and can require up to 21 constants (see Solid state physics).

The equation of motion

Newton’s second law for a small element gives Cauchy’s equation of motion, which holds for all continua – solids and fluids alike:

$$\rho \frac{D\vec{v}}{Dt} = \nabla \cdot \sigma + \rho \vec{g}$$

Insert the stress of a fluid, $\sigma_{ij} = -p\,\delta_{ij} + \eta(\partial_i v_j + \partial_j v_i)$, and one obtains the Navier–Stokes equations. Insert Hooke's law for a solid, and one obtains the Navier–Cauchy equation for the displacement:

$$\rho\,\frac{\partial^2 \vec{u}}{\partial t^2} = (\lambda + 2\mu)\,\nabla(\nabla \cdot \vec{u}) - \mu\,\nabla \times (\nabla \times \vec{u})$$

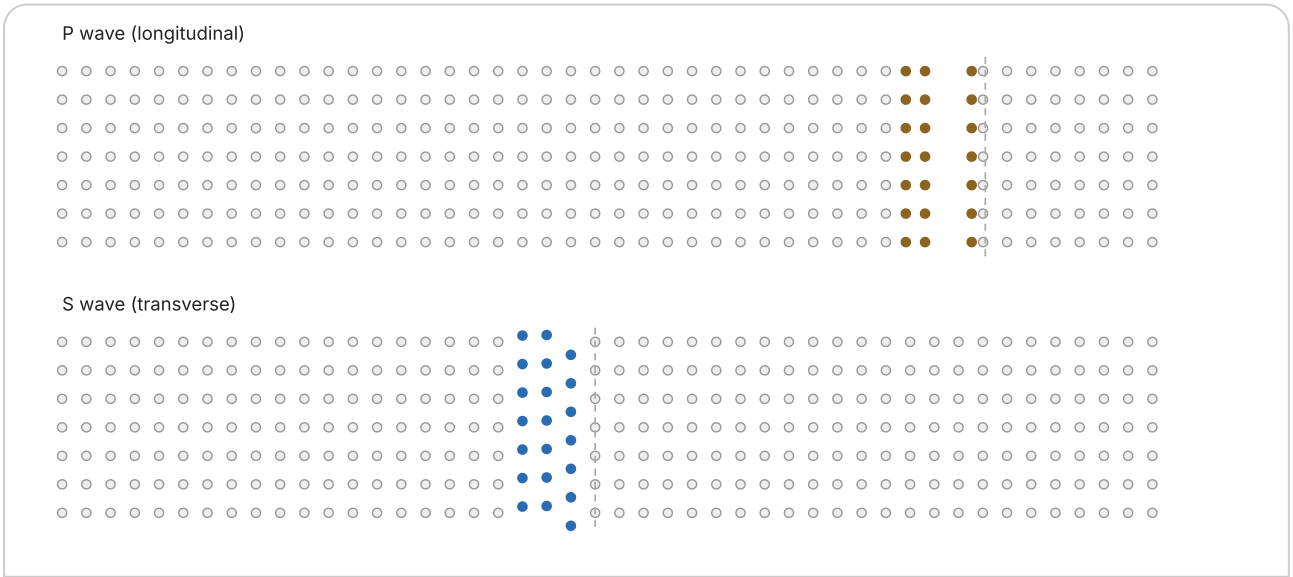
The difference between a solid and a fluid is thus whether the stress depends on the deformation or on the *rate* of deformation.

Elastic waves

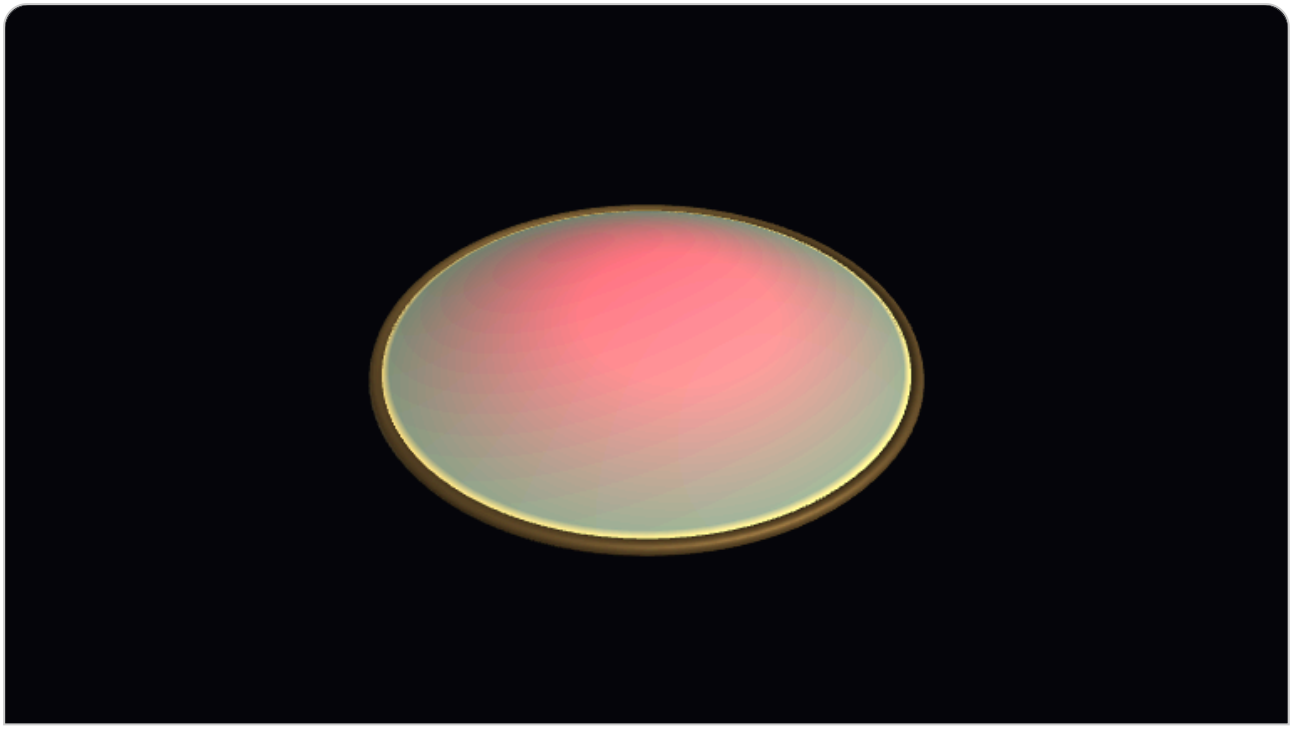
In 1829 Poisson showed that the Navier–Cauchy equation has two kinds of wave. The irrotational part ($\nabla \times \vec{u} = 0$) gives longitudinal pressure waves, the divergence-free part transverse shear waves:

$$v_P = \sqrt{\frac{\lambda + 2\mu}{\rho}} = \sqrt{\frac{K + \frac{4}{3}G}{\rho}}, \qquad v_S = \sqrt{\frac{\mu}{\rho}}, \qquad \frac{v_P}{v_S} = \sqrt{\frac{2(1-\nu)}{1-2\nu}}$$

The P wave is always the fastest and arrives first – hence the name (primary). The S wave (secondary) cannot pass through liquids, where $\mu = 0$. From the shadow of the S waves Richard Oldham and Beno Gutenberg concluded in 1906–1914 that the Earth's outer core is liquid, and in 1936 the Dane Inge Lehmann found the solid inner core from weak P waves that ought to have been in the shadow (see Earthquakes). There are also surface waves – Rayleigh waves (1885) and Love waves (1911) – which do the most damage in earthquakes.



Vibrations of a drumhead (3D)



Beams and buckling

A beam that is bent is stretched on one side and compressed on the other. Euler–Bernoulli beam theory (about 1750) gives the deflection $w(x)$ under a load per length $q(x)$:

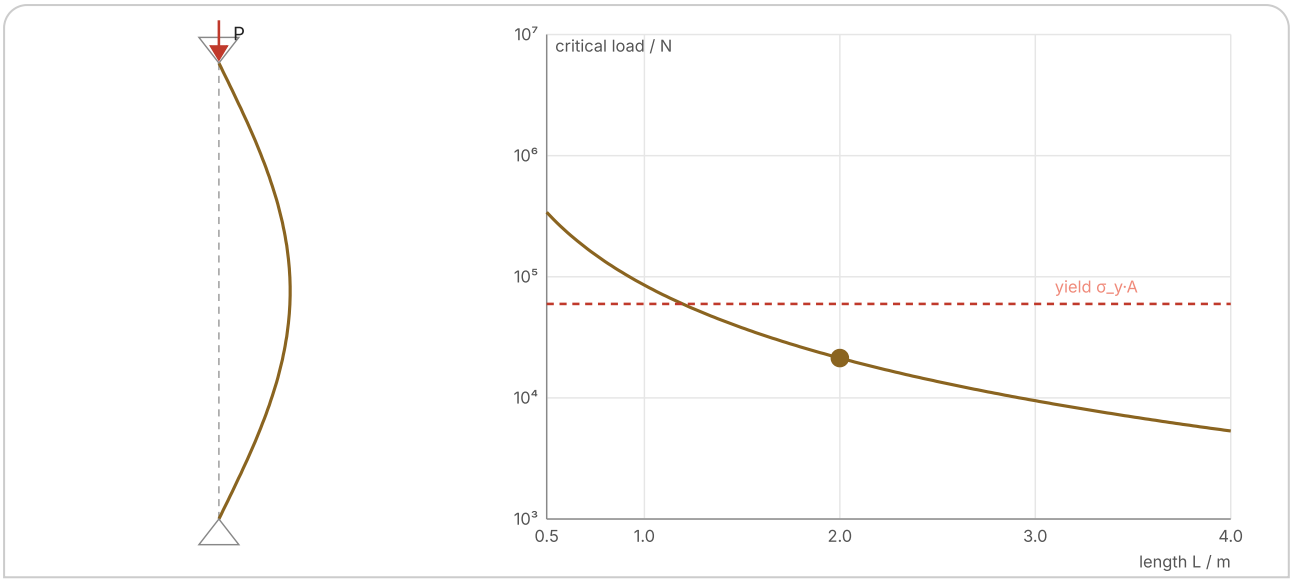
$$EI\,\frac{d^4w}{dx^4} = q(x), \qquad I = \int y^2\,dA$$

The second moment of area I grows as the cube of the depth – which is why beams are deep and I-shaped, and why a sheet of paper can carry more when folded. A cantilever of length L with an end load P deflects by $PL^3/3EI$. Galileo discussed the strength of beams in *Two New Sciences* (1638), but got the answer wrong.

A slender column under compression suddenly buckles sideways before the material fails. Euler derived the critical load in 1744:

$$P_{\text{cr}} = \frac{\pi^2 EI}{(KL)^2}$$

where K depends on how the ends are held. Buckling is a bifurcation: below P_{cr} the straight column is stable, above it it is unstable.



Plasticity and fracture

Above the yield stress a metal deforms plastically – it does not return to its original shape. This is due to dislocations sliding in the crystal lattice (Taylor, Orowan and Polanyi, 1934). When does yielding begin under a combined state of stress? Richard von Mises' criterion (1913) says: when the "equivalent stress" reaches the yield stress:

$$\sigma_{\mathrm{vM}} = \sqrt{\tfrac{1}{2}\left[(\sigma_1-\sigma_2)^2+(\sigma_2-\sigma_3)^2+(\sigma_3-\sigma_1)^2\right]} = \sigma_y$$

A hydrostatic pressure causes no yielding – only differences in stress count. Brittle materials such as glass fracture instead. In 1921 Alan Arnold Griffith explained why glass is a hundred times weaker than its theoretical strength: small cracks concentrate the stress, and a crack of length a grows when the energy released by opening it exceeds the surface energy, $\sigma_f \approx \sqrt{2E\gamma/\pi a}$. Fracture mechanics became essential when Liberty ships broke in two in the 1940s, and when the first Comet jet airliners crashed in 1954 because of fatigue cracks from their square windows.

Viscoelasticity and non-Newtonian materials

Many materials are both solid and liquid, depending on the time scale. Silly Putty bounces like a ball but flows out into a pancake over an hour. In 1867 Maxwell described this with a spring and a dashpot in series: the relaxation time $\tau = \eta/E$ separates elastic from viscous behaviour. The ratio of the relaxation time to the time of observation is called the Deborah number, after the song of the prophetess Deborah: "The mountains flowed before the Lord" – everything flows if you wait long enough. The pitch drop experiment in Queensland, started in 1927, has dripped only nine times.

Non-Newtonian fluids have a viscosity that depends on the shear rate: paint and ketchup get thinner when stirred (shear-thinning), cornflour in water goes hard when struck (shear-thickening – see Cornflour and water). Blood, lava, glaciers and the Earth's mantle are all viscoelastic; the mantle behaves as a solid for earthquake waves but flows like a liquid over millions of years and drives the continents.

Further reading

- L.D. Landau and E.M. Lifshitz: *Theory of Elasticity* (Course of Theoretical Physics vol. 7, 3rd ed. 1986)
- Y.C. Fung: *A First Course in Continuum Mechanics* (3rd ed., Prentice Hall 1994)
- J.E. Gordon: *The New Science of Strong Materials, or Why You Don't Fall Through the Floor* (Penguin 1968)
- J.E. Gordon: *Structures, or Why Things Don't Fall Down* (Penguin 1978)
- Richard Feynman: *The Feynman Lectures on Physics*, vol. II, ch. 38–39: "Elasticity" and "Elastic Materials"

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